
fourier series and Power Calculator

Designed By

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Department of Electrical and Electronics Engineering

Course No: EEE212

Course Name: Numerical Techniques Laboratory

Project

Fourier Series of Non-Sinusoidal Periodic Signal and Power Calculation

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Level: 2 **Term:** 1 **Section:** B1

Date of Submission: 20-02-2022

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1. Introduction

In the electrical system, it is very common to occur any unbalanced or distorted conditions where the system becomes non-sinusoidal and nonlinear. Actually, no electrical system can be made perfectly sinusoidal or absolutely linear. In that case, there arises many frequency components along with the fundamental component in the voltage and current waveform. But the problem is the usual wattmeter or varmeter cannot measure the real or reactive power. In that case, frequency domain analysis of a wave is necessary. Fourier series approximation and Fourier coefficients of the voltage and current signal help us to calculate power and power factor easily. That is why Fourier series approximation of electrical signal is necessary and in this report the theory, algorithm, main code, test cases, application, conclusion & user manual of the prepared project are discussed.

2. Theoretical Background

2.1. Determination of Fourier series

According to Baron Jean Baptiste Joseph Fourier, any periodic signal $f(t)$ with angular frequency ω can be expressed as,

$$f(t) = a_0 + a_1 \cos \omega t + a_2 \cos 2\omega t + \dots + a_n \cos n\omega t + \dots \\ + b_1 \sin \omega t + b_2 \sin 2\omega t + \dots + b_n \sin n\omega t + \dots$$

And the **coefficients** can be determined by

$$a_0 = \frac{1}{T} \int_{t_1}^{t_1 + T} f(t) dt$$

$$a_n = \frac{2}{T} \int_{t_1}^{t_1 + T} f(t) \cos n\omega t dt$$

$$b_n = \frac{2}{T} \int_{t_1}^{t_1 + T} f(t) \sin n\omega t dt$$

Average value of the signal $f(t)=a_0$

RMS value of the signal $f(t)$ with Fourier coefficients up to N sinusoidal components along with the fundamental,

$$F_{\text{rms}} = \frac{a_0^2}{4} + \frac{1}{2} \sum_{k=1}^{\infty} a_k^2 + b_k^2$$

Here the dc component is considered as ($a_{\text{dc}}=a_0/2$). If $a_{\text{dc}}=a_0$, then it will a_0^2 in the formula.

The signal $f(t)$ can also be expressed as:

$$f(t) = a_0 + c_1 \sin(\omega t + \theta_1) + c_2 \sin(2\omega t + \theta_2) + \dots$$

Where,

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta = \tan^{-1}\left(\frac{a_n}{b_n}\right)$$

2.2. Power Calculation

If the load where the voltage signal $f(t)$ is applied is known then the current signal can be found using the superposition principle in each of the phasor component and add them in time domain. Then we will have,

$$\begin{aligned} v(t) &= V_{\text{dc}} + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t - \theta_n) \\ i(t) &= I_{\text{dc}} + \sum_{m=1}^{\infty} I_m \cos(m\omega_0 t - \phi_m) \end{aligned}$$

Then the **real power** of the load will be,

$$P = V_{\text{dc}} I_{\text{dc}} + \frac{1}{2} \sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n)$$

Here, V_n and I_n are in rms form, that is why the factor (1/2) is required but if they were in peak/amplitude form then the factor (1/2) would not be required.

Similarly, **Reactive power** Q can be found by putting sine in place of cosine and neglecting the dc components. As dc waves do not have reactive power.

Then **Apparent power** S will be, $S = \sqrt{P^2 + Q^2}$

And the **Complex power** S will be, $S = P + Qi$

As the system is nonlinear, there is a measurement of the ratio of nonlinearity/distortion from pure sinusoid which is called THD. Two types of THD are measured, (i) THD_F and (ii) THD_R.

$$\text{THD}_F = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1} \quad \text{THD}_R = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{\sqrt{V_1^2 + V_2^2 + V_3^2 + \dots}} = \frac{\text{THD}_F}{\sqrt{1 + \text{THD}_F^2}}$$

For nonlinear system, two types of power factor are needed, (i) Displacement Power Factor (What we know as pf for sinusoids) and (ii) Distortion Power Factor (=1 for pure sinusoids as they have no distortion)

Displacement Power Factor (PFD) can be found from,

$$\text{power factor} = \frac{P, \text{ real power}}{|S|, \text{ apparent power}}$$

Distortion Power Factor (DPF) can be found from,

$$\text{distortion power factor} = \frac{1}{\sqrt{1 + \text{THD}_i^2}}$$

Then the **actual power factor** PF will be, $PF = PFD \times DPF$

$$PF = (\text{distortion factor})(\text{displacement factor})$$

3. Graphical User Interface (GUI)

3.1. Home Page

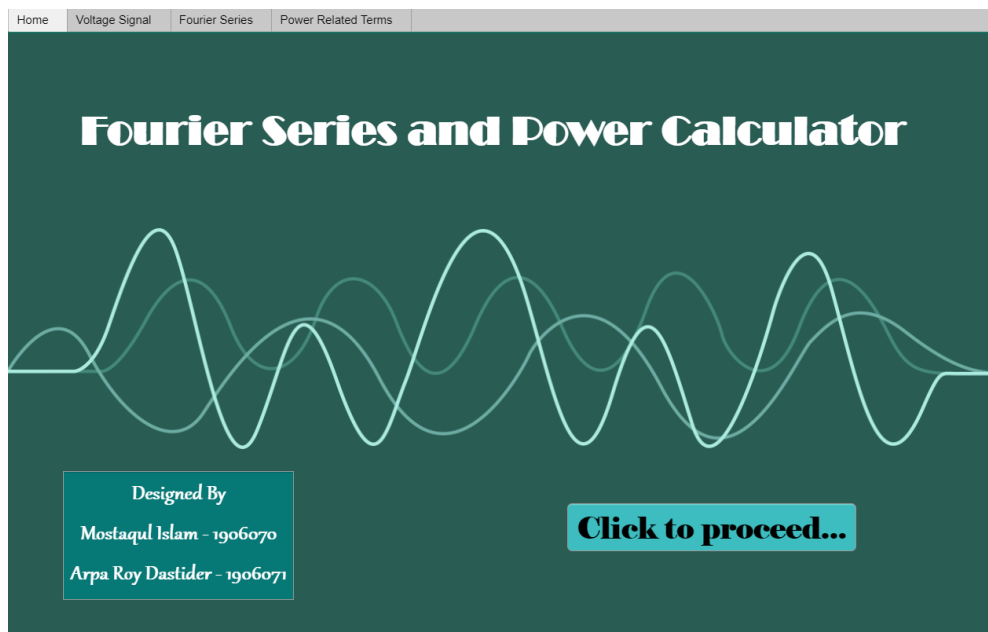


Fig-3.1: Home Page

At first, the user will open the home page which is just an introduction. User must click the button to proceed further and find desired outputs.

3.2. Voltage Signal

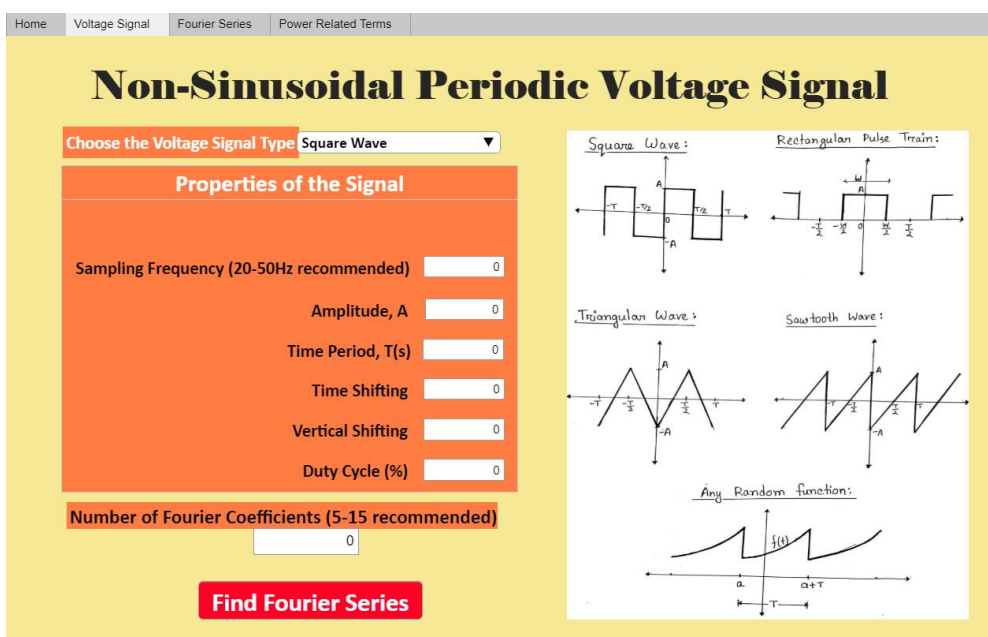


Fig-3.2: Page – 2 Voltage Signal

After pressing “click to proceed” button, this page will appear. In this page, the user will first select the signal type from the drop down menu called “Choose the voltage signal type” which he wants to analyze further. There are five options for the user.

- Square Wave**
- Rectangular Pulse Train**
- Triangular Wave**
- Swatooth Wave**
- Any Random Function**

According to the signal type, options for properties will change. Then the user will enter the properties of the signal. The properties are shown in the image on the right side. Finally the user will enter the number of Fourier coefficients he wants to determine. The process will complete when he presses the “Find Fourier Series” button.

3.3. Fourier Series

Fig-3.3: Page – 3 Fourier Series

When all the inputs are given and “Find Fourier Series” button is pressed, all the calculations occur according to the code. After finishing calculation, this page pops up automatically. In the

upper half of left side, the values of Fourier coefficients along with the average and rms value of the signal are printed automatically. In the bottom half of left side, the original signal and the Fourier approximation are plotted in the same figure. In the right half of this page, amplitude and phase spectrum corresponding to the angular frequencies are plotted in two separate figure. Lastly, the “Calculate Power Related Terms” button proceed the user to calculate power.

3.4. Power Related Terms

Fig-3.4: Page – 4 Power Related Terms

If the user wants to calculate power when the voltage signal is applied across a load and presses the button “Calculate Power Related Terms”, this page opens. First of all, the user will choose the load type from the dropdown menu. There are six options.

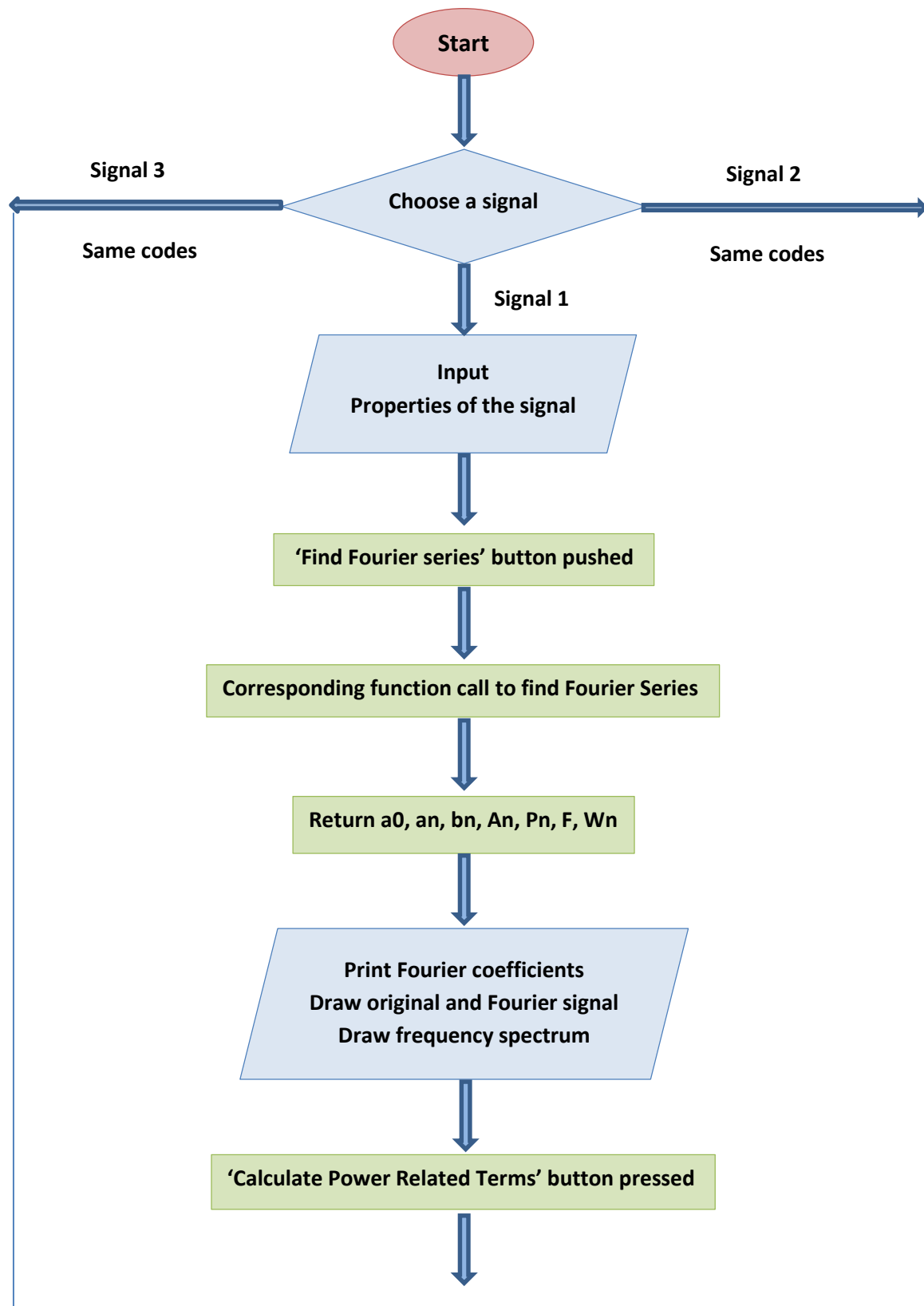
When the load type is chosen, the user will enter the values of resistance, inductance and capacitance according to the type. After clicking “Calculate” button, all the power related terms pop up in the right panel. The user can calculate power as many time he wants.

3.5. User Manual

- ❖ Sample frequency should not be less than 20 and more than 50. Recommended/Rated sample frequency is 35.
- ❖ The number of Fourier coefficients should not be less than 5 and more than 15. Recommended/Rated number is 10.
- ❖ If random function is to be given as input, then it is recommended to write the whole function inside a first bracket.
- ❖ Any practical load cannot be perfectly inductive or capacitive. So, it is better not to choose a pure inductor/a pure capacitor as a load.
- ❖ User should wait for a while until the fourier calculation is performed.
- ❖ User should give continuous function as input in the given interval.
- ❖ User can frequently change the pages and compare input and output using the tab option. User can also run the whole program as many times as he wants by switching to the input page.

Home	Voltage Signal	Fourier Series	Power Related Terms
------	----------------	----------------	---------------------

4. Flow Chart of the Program



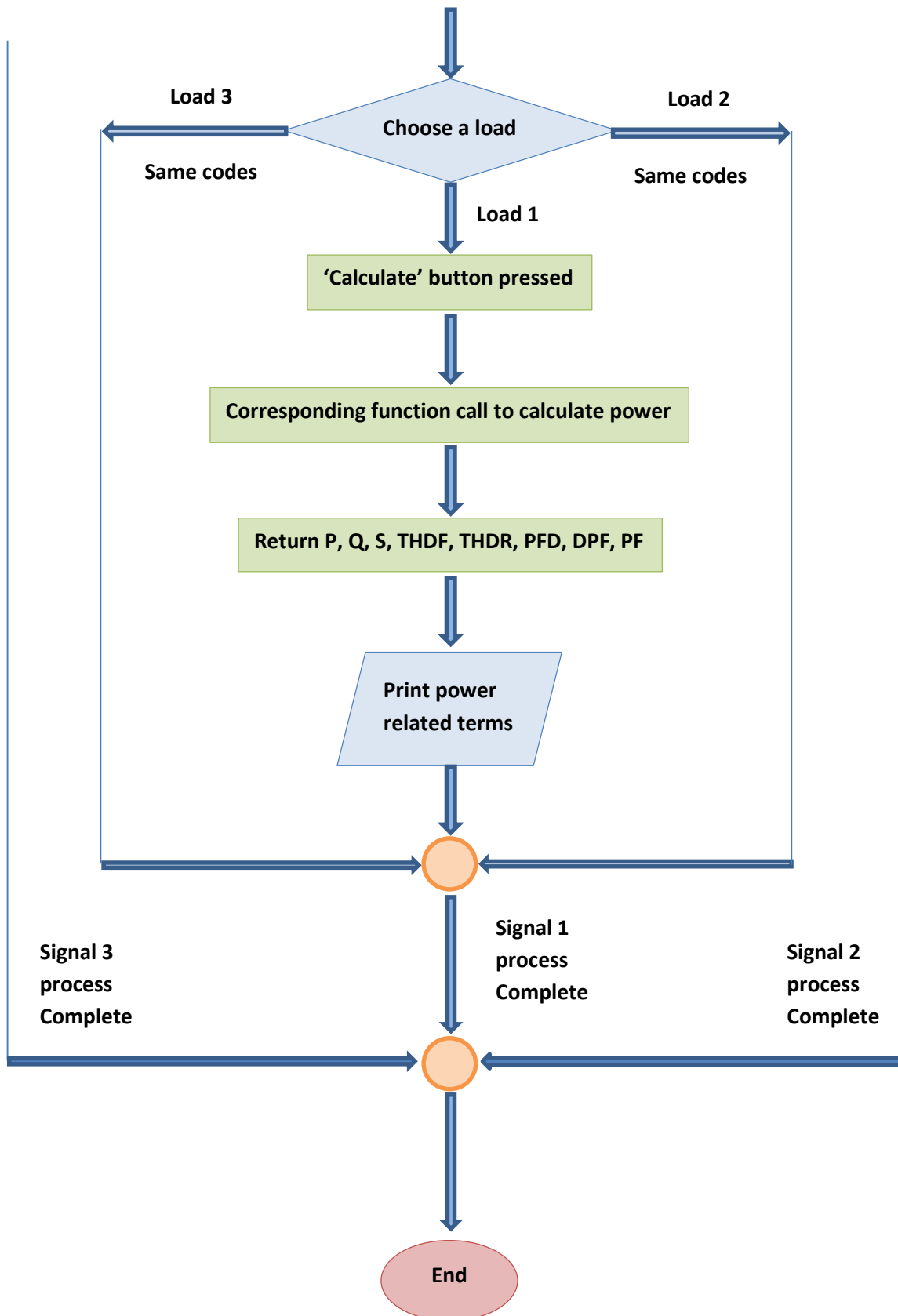


Fig-4.1: Flow chart of the program

5. Main Code:

First of all, properties of the signal to be analyzed are taken from the user as inputs. The inputs are saved in the variables as f – function; f_s – sampling frequency; T – Time Period; D_c – Duty Cycle; amp - Amplitude; x_s – Time Shifting; v_s – Vertical Shifting; W – Width of rectangular pulse; a – starting point of the central period; N – Number of fourier coefficients.

5.1. Find Fourier series:

When this button is pushed, the main code runs to find the coefficients according to the selected type of the signal.

(i) Square Wave:

```
t1=(Dc/100)*T;

w0=(2*pi)/T;
k=-((5*T/2)-1/fs):1/fs:((5*T/2)-1/fs);
syms t %t is systematic variable now for integrstion
n=1:N;
a0=(1/T)*(int(amp+vs,t,xs,xs+t1)+int(-amp+vs,t,xs+t1,xs+T));
an=(2/T)*(int((amp+vs)*cos(n*w0*(t-xs)),t,xs,xs+t1)+int((-amp+vs)*cos(n*w0*(t-xs)),t,xs+t1,xs+T));
bn=(2/T)*(int((amp+vs)*sin(n*w0*(t-xs)),t,xs,xs+t1)+int((-amp+vs)*sin(n*w0*(t-xs)),t,xs+t1,xs+T));
F=a0;
for i=1:N
    F=F+(an(i)*cos(i*w0*(k-xs)))+(bn(i)*sin(i*w0*(k-xs)));
end
f=1/T;
Wn=zeros(1,N+1);
An=zeros(1,N+1);
Pn=zeros(1,N+1);
Wn(1)=0;
An(1)=a0;
Pn(1)=0;
for i=2:N+1
    Wn(i)=(2*pi*(i-1)*f);
    An(i)=sqrt(an(i-1)^2+bn(i-1)^2);
    if an(i-1)~=0
        Pn(i)= -((atan(bn(i-1)/an(i-1)))*(180/pi));
    elseif (an(i-1)==0 && bn(i-1)==0)
        Pn(i)= 0;
    else
        Pn(i)= -90;
    end
end
end
```

(ii) Rectangular Pulse Train:

```
w0=(2*pi)/T;
k=-(2*T+(W/2)-xs)-1/fs:1/fs:(2*T+(W/2)+xs)+1/fs;
syms t %t is systematic variable now for integrstion
n=1:N;
a0=(1/T)*((int(amp+vs,t,-(W/2-xs),W/2+xs))+int(0+vs,t,(W/2+xs),(T-
W/2+xs))));
an=(2/T)*((int((amp+vs)*cos(n*w0*t),t,-(W/2-
xs),W/2+xs))+int((0+vs)*cos(n*w0*t),t,(W/2+xs),(T-W/2+xs))));
bn=(2/T)*((int((amp+vs)*sin(n*w0*t),t,-(W/2-
xs),W/2+xs))+int((0+vs)*sin(n*w0*t),t,(W/2+xs),(T-W/2+xs))));
F=a0;
for i=1:N
    F=F+(an(i)*cos(i*w0*k))+(bn(i)*sin(i*w0*k));
end
f=1/T;
Wn=zeros(1,N+1);
An=zeros(1,N+1);
Pn=zeros(1,N+1);
Wn(1)=0;
An(1)=a0;
Pn(1)=0;
for i=2:N+1
    Wn(i)=(2*pi*(i-1)*f);
    An(i)=sqrt(an(i-1)^2+bn(i-1)^2);
    if an(i-1)~=0
        Pn(i)= -((atan(bn(i-1)/an(i-1)))*(180/pi));
    elseif (an(i-1)==0 && bn(i-1)==0)
        Pn(i)= 0;
    else
        Pn(i)= -90;
    end
end
end
```

(iii) Triangular Wave:

```
w0=(2*pi)/T;
k=-((5*T/2)-1/fs):1/fs:((5*T/2)-1/fs);
syms t %t is systematic variable now for integrstion
%N=10;
m1=(4*amp)/T;
m2=-m1;
n=1:N;
f1=@(x) (m1*(x-xs-T/4))+vs ;
f2=@(x) (m2*(x-xs-((3*T)/4)))+vs ;
a0=(1/T)*((int(f1(t),t,xs,xs+T/2))+int(f2(t),t,xs+T/2,xs+T));
an=(2/T)*((int(f1(t)*cos(n*w0*(t-xs)),t,xs,xs+T/2))+int(f2(t)*cos(n*w0*(t-
xs)),t,xs+T/2,xs+T));
bn=(2/T)*((int(f1(t)*sin(n*w0*(t-xs)),t,xs,xs+T/2))+int(f2(t)*sin(n*w0*(t-
xs)),t,xs+T/2,xs+T));
F=a0;
for i=1:N
    F=F+(an(i)*cos(i*w0*(k-xs)))+(bn(i)*sin(i*w0*(k-xs)));
end
```

```

f=1/T;

Wn=zeros(1,N+1);
An=zeros(1,N+1);
Pn=zeros(1,N+1);
Wn(1)=0;
An(1)=a0;
Pn(1)=0;
for i=2:N+1
Wn(i)=(2*pi*(i-1)*f);
An(i)=sqrt(an(i-1)^2+bn(i-1)^2);
    if an(i-1)~=0
        Pn(i)= -((atan(bn(i-1)/an(i-1)))*(180/pi));
    elseif (an(i-1)==0 && bn(i-1)==0)
        Pn(i)= 0;
    else
        Pn(i)= -90;
    end
end
end

```

(iv) Sawtooth Wave:

```

w0=(2*pi)/T;
k=-((5*T/2)-1/fs):1/fs:((5*T/2)-1/fs);
syms t %t is systematic variable now for integrstion
n=1:N;
f=@(x) (((2*amp)/T)*(x-xs-T/2))+vs ;
a0=(1/T)*(int(f(t),t,xs,xs+T));
an=(2/T)*(int(f(t)*cos(n*w0*(t-xs)),t,xs,xs+T));
bn=(2/T)*(int(f(t)*sin(n*w0*(t-xs)),t,xs,xs+T));
F=a0;
for i=1:N
    F=F+(an(i)*cos(i*w0*(k-xs)))+(bn(i)*sin(i*w0*(k-xs)));
end
f=1/T;
Wn=zeros(1,N+1);
An=zeros(1,N+1);
Pn=zeros(1,N+1);
Wn(1)=0;
An(1)=a0;
Pn(1)=0;
for i=2:N+1
Wn(i)=(2*pi*(i-1)*f);
An(i)=sqrt(an(i-1)^2+bn(i-1)^2);
    if an(i-1)~=0
        Pn(i)= -((atan(bn(i-1)/an(i-1)))*(180/pi));
    elseif (an(i-1)==0 && bn(i-1)==0)
        Pn(i)= 0;
    else
        Pn(i)= -90;
    end
end
end

```

(v) Any Random Function:

```
w0=(2*pi)/T;
k=((a-2*T)+xs) : 1/fs : ((a+3*T)+xs);
g=f;
syms t %t is systematic variable now for integrstion
n=1:N;
a0=(1/T)*(int( f(t)+vs ,t, a, a+T) );
an=(2/T)*(int( (f(t)+vs)*cos(n*w0*(t)) ,t, a, a+T) );
bn=(2/T)*(int( (f(t)+vs)*sin(n*w0*(t)) , t, a, a+T) );
F=a0;
for i=1:N
    F=F+(an(i)*cos(i*w0*(k-xs) ))+(bn(i)*sin(i*w0*(k-xs) ));
end
f=1/T;
Wn=zeros(1,N+1);
An=zeros(1,N+1);
Pn=zeros(1,N+1);
Wn(1)=0;
An(1)=a0;
Pn(1)=0;
for i=2:N+1
    Wn(i)=(2*pi*(i-1)*f);
    An(i)=sqrt(an(i-1)^2+bn(i-1)^2);
    if an(i-1)~=0
        Pn(i)= -((atan(bn(i-1)/an(i-1)))*(180/pi));
    elseif (an(i-1)==0 && bn(i-1)==0)
        Pn(i)= 0;
    else
        Pn(i)= -90;
    end
end
end
```

The graphs of fourier approximation, amplitude spectrum and phase spectrum are plotted using the values got from aforementioned codes. The average and rms value are determined by the following lines:

```
ms=(app.a0)^2;
for i=1:app.N
    ms=ms+app.an(i)^2+app.bn(i)^2;
end
rms=sqrt(ms);
```

5.2. Power Calculation:

If the voltage signal is applied across a load, power related terms can be calculated according to the load type. The process is more or less same for all types of load. Code for combined R, L, C in series is given below:

```

b0=a0/Inf;
w0=(2*pi)/T;
P=0;
Q=0;
THDF=0;
for j=2:N+1
    Zeq= R+( (((j-1)*w0*L)- (1/( (j-1)*w0*C) ) ) *i );% real and imag
    Z=abs(Zeq);
    ang=(angle(Zeq))*(180/pi);

    Bn(j)= An(j)/Z;
    Hn(j)= Pn(j)-ang;
end
Bn(1)=b0;
Hn(1)=Pn(1);
for j=1:N+1
    P=P+ An(j)*Bn(j)*cos( ((Pn(j)-Hn(j))*(pi/180)) );
    Q=Q+ An(j)*Bn(j)*sin( ((Pn(j)-Hn(j))*(pi/180)) );
end
S=sqrt( (P^2)+ (Q^2) );
for j=3:N+1
    THDF=THDF+( (An(j))^2);
end
THDF= (sqrt(THDF) )/An(2);
THDR= ( THDF )/ ( sqrt(1+ (THDF)^2 ) );

PFD=(P/S);%PFD=Displacement Power Factor
DPF= ( 1 )/ ( sqrt(1+ (THDF)^2 ) );%DPF=Distortion Power Factor
PF=(PFD)*(DPF);%PF=True Power Factor

```

6. Test Cases

Example-1:

Input is a square wave:

Corresponding Output:

Choose the Voltage Signal Type
Square Wave

Properties of the Signal

Sampling Frequency (20-50Hz recommended)
20

Amplitude, A
5

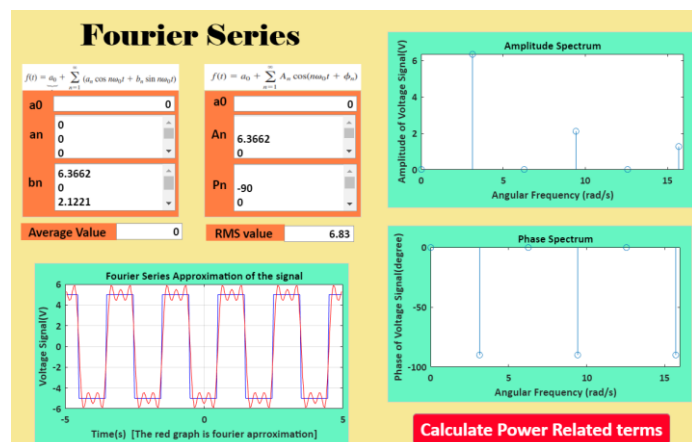
Time Period, T(s)
2

Time Shifting
0.5

Vertical Shifting
0

Duty Cycle (%)
50

Number of Fourier Coefficients (5-15 recommended)
5



Load for same signal:

Choose load type **R,L,C in series** ▼

Load Values

Resistance (ohm)

Inductance (Henry)

Capacitance (Farad)

Power:

Power Factor Type **Leading**

Real/Average Power, P **0.00141993 W**

Reactive Power, Q **-0.01937451 VAR**

Apparent Power, S **0.01942647 VA**

Complex Power, S **0.00141993-0.01937451i VA**

Total Harmonic Distortion (fundamental frequency), THDF **0.38873013**

Total Harmonic Distortion (root mean square), THDR **0.36231772**

Displacement power factor, PFD **0.07309258 (Leading)**

Distortion power factor, DPF **0.93205465**

True Power Factor, PF **0.06812628**

Example-2:

Input is a rectangular pulse train: Output:

Choose the Voltage Signal Type **Rectangular Pulse Train** ▼

Properties of the Signal

Sampling Frequency (20-50Hz recommended)

Amplitude, A

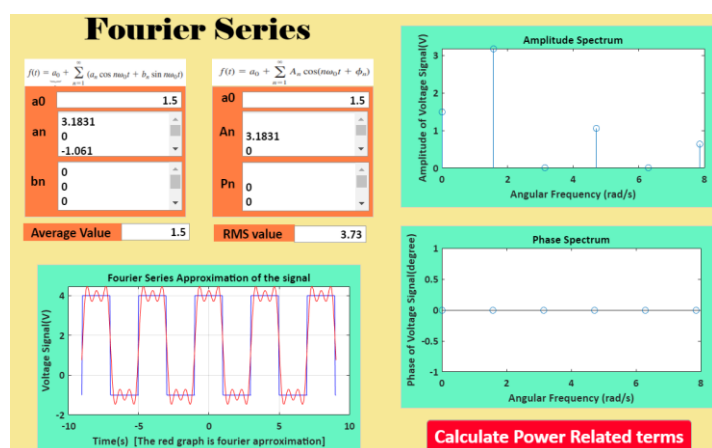
Time Period, T(s)

Time Shifting

Vertical Shifting

Width, W (< Time period)

Number of Fourier Coefficients (5-15 recommended)



Load for same signal:

Choose load type **R,C in series** ▼

Load Values

Resistance (ohm)

Capacitance (Farad)

Power:

Power Factor Type **Leading**

Real/Average Power, P **0.00183051 W**

Reactive Power, Q **-0.01100160 VAR**

Apparent Power, S **0.01115285 VA**

Complex Power, S **0.00183051-0.01100160i VA**

Total Harmonic Distortion (fundamental frequency), THDF **0.38873013**

Total Harmonic Distortion (root mean square), THDR **0.36231772**

Displacement power factor, PFD **0.16412955 (Leading)**

Distortion power factor, DPF **0.93205465**

True Power Factor, PF **0.15297771**

Example-3:

Input is a Triangular Wave:

Output:

Choose the Voltage Signal Type **Triangular Wave**

Properties of the Signal

Sampling Frequency (20-50Hz recommended)

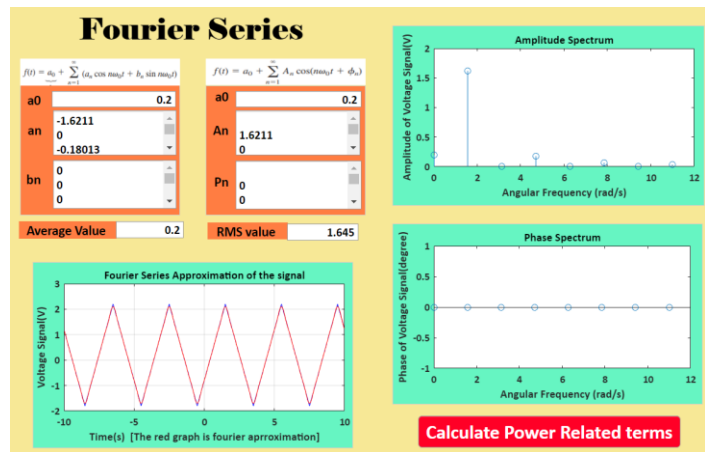
Amplitude, A

Time Period, T(s)

Time Shifting

Vertical Shifting

Number of Fourier Coefficients (5-15 recommended)



Load for same signal:

Power:

Choose load type **R,L in series**

Load Values

Resistance (ohm)

Inductance (Henry)

Power Factor Type	Lagging
Real/Average Power, P	0.01176448 W
Reactive Power, Q	0.00001636 VAR
Apparent Power, S	0.01176449 VA
Complex Power, S	0.01176448+0.00001636i VA
Total Harmonic Distortion (fundamental frequency), THDF	0.11984228
Total Harmonic Distortion (root mean square), THDR	0.11899084
Displacement power factor, PFD	0.99999903 (Lagging)
Distortion power factor, DPF	0.99289535
True Power Factor, PF	0.99289439

Example-4:

Input is a Sawtooth Wave:

Output:

Choose the Voltage Signal Type **Sawtooth Wave**

Properties of the Signal

Sampling Frequency (20-50Hz recommended)

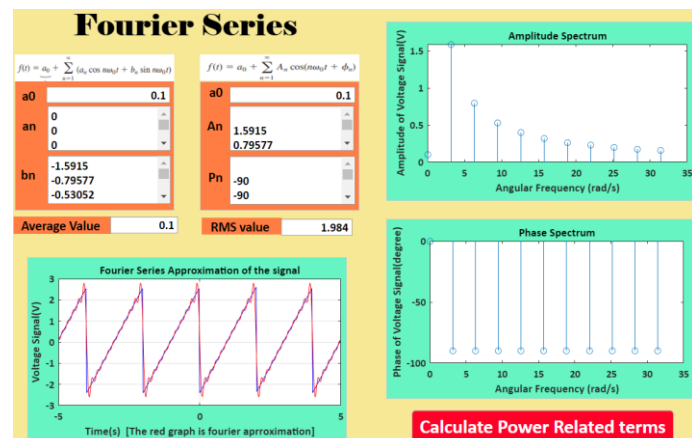
Amplitude, A

Time Period, T(s)

Time Shifting

Vertical Shifting

Number of Fourier Coefficients (5-15 recommended)



Load for same signal:

Choose load type **Pure Resistor (R)** ▼

Load Values

Resistance (ohm)

Power:

Power Factor Type **Unity**

Real/Average Power, P **0.01967804 W**

Reactive Power, Q **0.00000000 VAR**

Apparent Power, S **0.01967804 VA**

Complex Power, S **0.01967804 VA**

Total Harmonic Distortion (fundamental frequency), THDF **0.74146324**

Total Harmonic Distortion (root mean square), THDR **0.59560223**

Displacement power factor, PFD **1.00000000 (Unity)**

Distortion power factor, DPF **0.80327952**

True Power Factor, PF **0.80327952**

Example-5:

Input is a function:

Choose the Voltage Signal Type **Any Random Function** ▼

Properties of the Signal

Function, f(t) **(t^3)**

Sampling Frequency (20-50Hz recommended) **35**

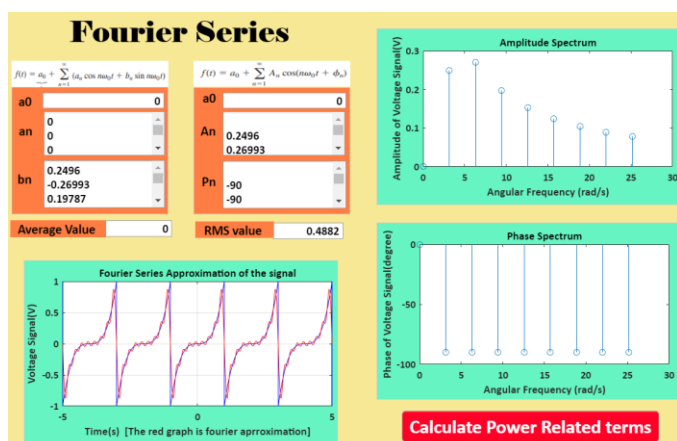
Starting point of the central period, a **-1**

Time Period, T(s) **2**

Time Shifting **0**

Vertical Shifting **0**

Number of Fourier Coefficients (5-15 recommended) **8**



Load for same signal:

Choose load type **Pure Capacitor (C)** ▼

Load Values

Capacitance (Farad)

Power:

Power Factor Type **Leading**

Real/Average Power, P **0.00000000 W**

Reactive Power, Q **-0.00098615 VAR**

Apparent Power, S **0.00098615 VA**

Complex Power, S **0.00000000-0.00098615i VA**

Total Harmonic Distortion (fundamental frequency), THDF **1.68102536**

Total Harmonic Distortion (root mean square), THDR **0.85943000**

Displacement power factor, PFD **0.00000000 (Leading)**

Distortion power factor, DPF **0.51125344**

True Power Factor, PF **0.00000000**

7. Applications

- ❖ Communication engineering deals with various types of signals including non-sinusoidal periodic signals. Those can be expressed as a summation of sinusoidal terms using Fourier series.
- ❖ Fourier series allows to extract frequency components of a signal. Information encoded into various frequencies can be pulled out with this feature.
- ❖ Noise cancellation of a signal can be done by using frequency spectrum.
- ❖ The Fourier approximations and coefficients help to calculate power and power factor easily.
- ❖ Power calculation of signals with various frequency components are not possible with normal wattmeter and varmeter. Fourier series makes it possible to calculate the power related terms.

8. Shortcomings

- ❖ In this program, the user cannot draw the input function arbitrarily. The user should have to give the function of the input voltage signal.
- ❖ The user should give function as input which will be continuous in the given interval.
- ❖ There are two types of facilities for input signal. First one is some built in signals and the second one is random signal. In both case, if the signal has some discontinuities, then fourier approximation becomes very poor at the point of discontinuities which is about 9% overshoot.
- ❖ The user has to adjust the load according to the given menu. Any combination of R, L, C cannot be produced in this program.
- ❖ Fourier approximation takes a very long time to be executed if user gives input for $N=40$ or $N=50$. The code will execute but it will require huge time.
- ❖ Sample frequency should be at least twice of the input signal frequency. But it should not be more than 50. As the sample frequency is higher, it takes more time to show the output

9. Conclusion

There is no software without any glitch or bug or error. That is why a program should be updated time to time. Our program can also be improved to correct the shortcomings stated before maybe in the future:

- The program can be more user friendly by giving the scope to draw any random function to be approximated by Fourier series.
- The program can be made fast & efficient so that it can approximate 100 coefficients very fast.
- The program can be added to any circuit simulator software to take any input voltage and load.
- Fourier series can also be generalized by Fourier transform.
- The program can be linked with a hardware/machine to make a General Power Meter which will calculate power, power factor of any electrical system.

That's why Fourier series approximation was chosen as a project for EEE 212 course. Apart from the shortcomings, the user interface of this project has been made as much user friendly as possible and it is hoped that this project will help users in calculating desired Fourier series and power.